CREATIVE THINKING & INNOVATION

Unit-1 Introduction

Creative thinking

Creativity: Bringing into existence an idea that is new to you

Innovation: The practical application of creative ideas

Creative thinking: an innate talent that you were born with and set of skills that can be learned developed and utilized in daily problem solving.

Creative solutions have three attributes:

1) It is new
2) It is useful, in that it solves the problem.
3) It is feasible.

Types of thinking

Convergent thinking: narrowing down a problem to find the one probable solution.

Divergent thinking: open minded thinking; individual thinks of various answers to single question as per experience.

Strategies for creative thinking

1. Cultivate habit of wider reading & develop habit of asking questions.
2. Generate as many ideas as possible for particular problems; think of alternatives; never accept the 1^{st} idea; let more come & then select the best.
3. Brainstorming: - let you mind think freely and let the ideas come, do not start evaluating their worth already.
4. Practice making unusual association & analogies. Let the conflicting thought co-exist.
5. Dwell in ambiguity. Let your ideas stay in mind for a while, you may come up with something enlightening.
6. Indulge in activities which bring forward your originality & get feedback from outsiders ie. Those not involved.
7. Try to predict events and analyse what change might bring in case of failure, cope with it, be self-confident & positive.

Innovation : Creative ideas when transformed into product or process is called an innovation

Types of innovation

1. Business model innovation involves changing the way business is done in terms of capturing value eg. HP vs Dell
2. Process innovation involves the implementation of new or significantly improves production or delivery method
3. Product innovation involves the introduction of new good or service that is new or substantially improved. This might include improvements in functional characteristics, technical abilities, ease of use or any other dimension.
4. Service innovation is similar to product innovation except that innovation relates to service rather than to products.

Blocks to creativity

Mental obstacles that constrain the way the problems is defined and limit the number of alternative solutions thought to be relevant.

1. Self-Doubt or Fear of Failure : One of the most common reasons people don't want to engage in creative thinking is that they are afraid to fail. This fear can be linked to the feeling of being judged by our peers or even ourselves.

2. No curiosity or non-inquisitiveness : Not asking questions , Sometimes the inability to solve problems results from reticence to ask questions, to obtain information, or to search for data.

3. Over reliance on Logic : Our thought processes, social interactions, and final decisions are all governed by logic. However, keep in mind that not everything has a logical explanation and that you may occasionally need to abandon logic in favour of a more imaginative approach to problem-solving.

4. Distinguishing figure from ground: Not filtering out irrevalent information or finding needed information. The inability to separate the important from un-important and to appropriately compress problems.

5. Artificial constraints: Defining boundaries of problems too narrowly. People assume that some problems definitions or alternative solutions are off limits, so they ignore them.

6. Vertical thinking : Defining problem in only one way without considering alternative views. Lateral thinkers on other hand generate alternative ways of viewing problem and produce multiple definitions.

7. One thinking language: using only one language (words) to define and assess the problem. Disregarding other languages such as non-verbal or symbolic languages.

Factors that influence creative design

1. Freedom: The level of independence affects creativity. If a person in business get the freedom to do something new without permission or fear of failure such person becomes creative.

2. Resources: Available resources also affects creativity. Sufficient resources (human resources, capital, machines, raw material and methods) are encouraged by creative activities.

3. Pressure: The level of pressure at work also affect creativity. A certain level of pressure to work expands creativity. When there is lot of work to be done in a short period of time. People have to think of new and creative ways to do it within allotted time.

4. Organizational culture : Organizational culture encourages creativity through fear, creative conclusion of ideas, identification encouragement of creative work, mechanisms for developing new ideas.

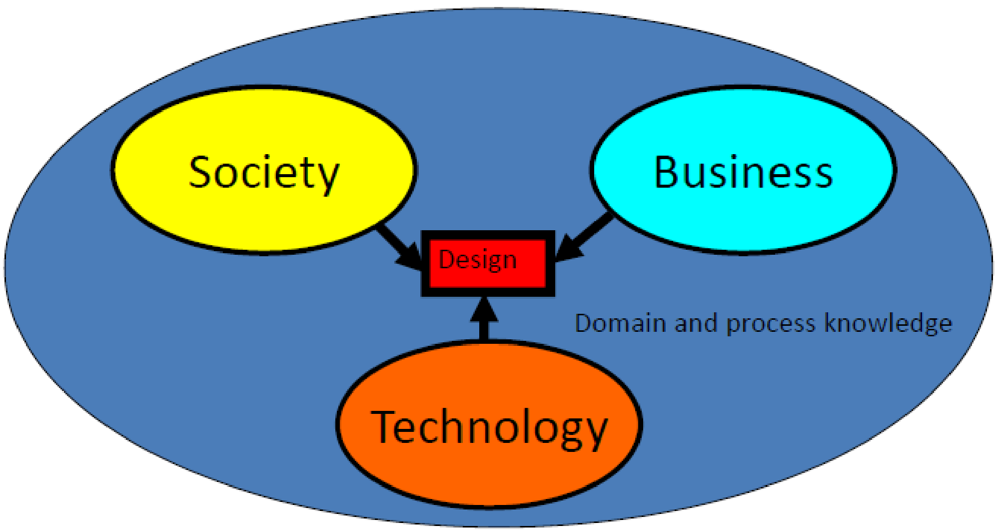
5. Encouragement of creativity : Creativity is also influenced by level of encouragement of creativity from working group.

Influence of society, technology and business on creativity

The influence of society, technology, and business on creativity is multifaceted and interdependent. Society provides the cultural and social context, technology offers tools and platforms for creation, and business supplies the economic framework and incentives. Together, they create a dynamic environment that shapes the possibilities and directions of human creativity. The synergy between society, business, and technology shapes and propels new design creativity.

Eg.

- Whose perception plays an important role
  - The same car may be stylish to one and boring to another
  - One may find it "cheap", another may find it "expensive"
- Where it is perceived plays an important role
  - Ambassador car popular in parts of India (with bad roads) due to stability
  - In other parts, it is considered too heavy, slow, inefficient…
- When it is perceived plays an important role
  - The same car that was stylish become boring with time for the same person



Engineering design and creative design

Engineering design and creative design are two distinct but related processes that serve different purposes. Here are some key differences between the two:

Purpose:
- **Engineering Design:** Engineering design focuses on creating solutions that meet specific technical requirements and functional criteria. The primary goal is to develop products, systems, or structures that are safe, efficient, and cost-effective.
- **Creative Design:** Creative design is more focused on aesthetics, innovation, and user experience. It often involves creating visually appealing or emotionally engaging products, such as graphic design, fashion design, or industrial design.

Constraints:
- **Engineering Design:** Engineering design is bound by technical constraints such as material properties, structural integrity, safety standards, and manufacturing processes. Engineers must ensure that their designs are feasible and can be implemented within constraints.
- **Creative Design:** Creative design is less constrained by technical limitations and often involves pushing boundaries and exploring new ideas. Creativity and innovation are key drivers in creative design processes.

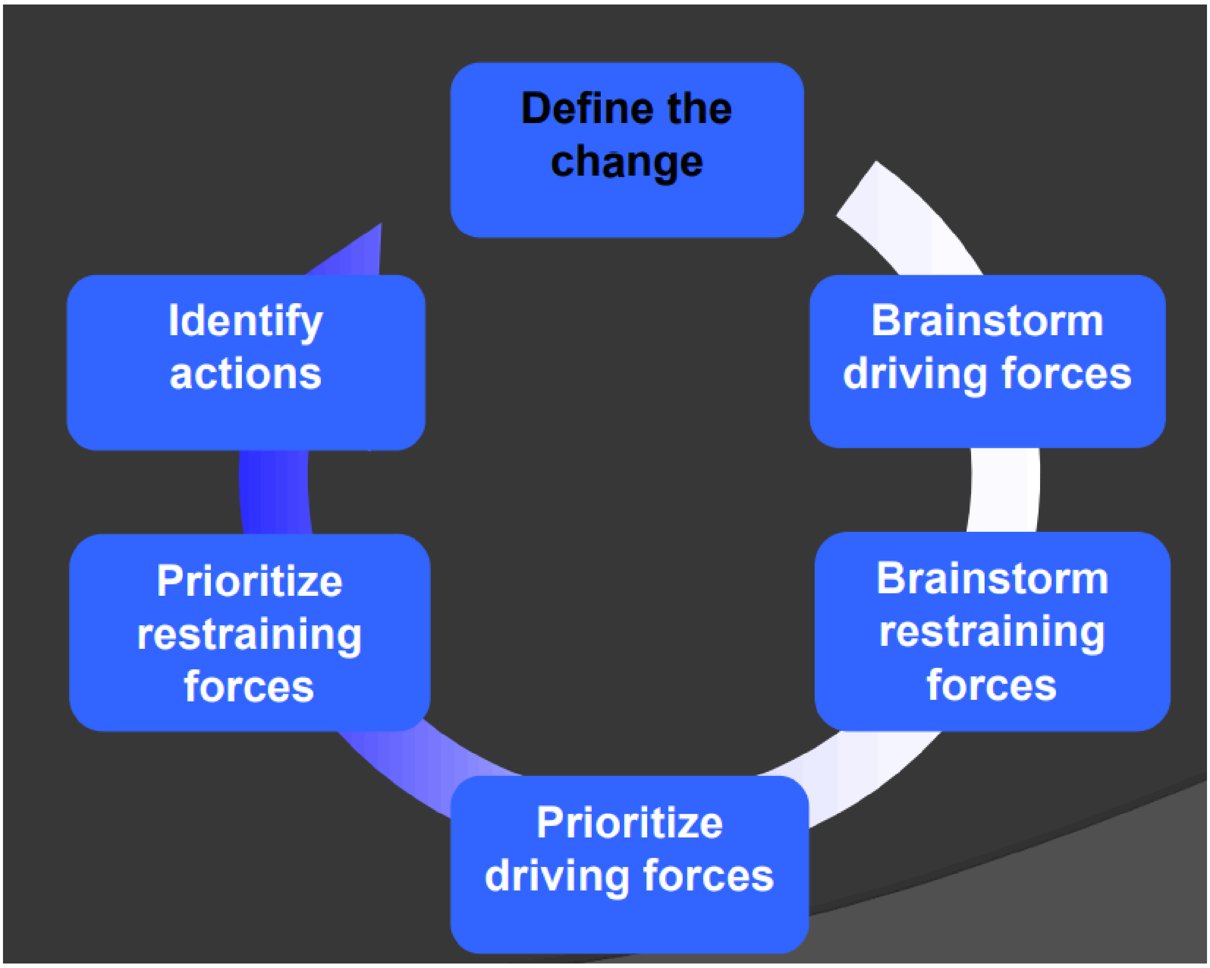
Problem-solving approach:
- **Engineering Design:** Engineering design typically follows a systematic problem-solving approach, starting with defining the problem, analyzing requirements, generating solutions, and evaluating alternatives. The emphasis is on practicality, efficiency, and functionality.
- **Creative Design:** Creative design often involves a more iterative and exploratory approach, where designers experiment with different ideas, styles, and concepts to arrive at innovative solutions. The focus is on originality, artistic expression, and emotional impact.

Evaluation criteria:
- **Engineering Design:** Engineering designs are typically evaluated based on criteria such as functionality, reliability, efficiency, safety, and cost-effectiveness. The success of an engineering design is often measured by how well it meets these criteria.
- **Creative Design:** Creative designs are evaluated based on criteria such as aesthetics, originality, usability, emotional appeal, and marketability. The success of a creative design is often judged by its impact on users and its ability to evoke a desired response.

Force Field Analysis

Force field analysis is a problem-solving tool used to help change occur. Force field analysis is the exercise of identifying the driving and restraining forces that surround a proposed change. Working through this process of identifying forces encourages creative thinking by forcing a team to think together about the aspects of the desired change. The exercise also encourages the team to agree on the priority forces. This agreement provides a starting point for action.

Force field analysis is used by teams when trying out their improvement theories (hypotheses). It is often used just after the team has generated improvement theories using the funnelling of data concept in which the nominal group technique has been applied. It is a powerful tool and can be used to help any desired change occur.



First, the forces on both sides must be identified. Then they must be weighed in terms of the amount of force they exert. When we can see more clearly what these various forces are and how significant (or strong) they are, there is a better chance of bringing change in the direction we seek.

Steps in Force Field Analysis

1. Identify problem - describe in writing the change desired.
2. Define problem in terms of:
   a) Present situation
   b) Situation you desire to see when problem is solved
3. List forces working for and against change.
4. Underline the most important forces/give a weighing.
5. For each restraining force list actions you could take to reduce/eliminate that force.
6. For each driving force list actions which would increase that force.
7. Determine most promising steps you could take in sequence.
8. Re-examine your steps for resources required and omit steps which do not help achieve your goal.

Force Field Analysis	
Desired Change:	
Driving Forces (+)	**Restraining Forces (-)**
ACTIONS: **1.** **2.** **3.**	

<u>Technology Push</u>

Technology Push is when research and development in new technology, drives the development of new products. Technology Push usually does not involve market research. It tends to start with a company developing an innovative technology and applying it to a product. The company then markets the product.

Eg. Touch screen technology

Touch Screen technology appeared as published research by E.A. Johnson at the Royal Radar

Establishment UK, in the mid 1960s. The technology began to attract research and development funding. In the 1980s, Hewlett Packard introduced a touch screen computer. 1993 hand writing recognition introduced - Apple's Newton PDA. 1996, Palm introduced its Pilot Series. Touch screen technology now seen in smart phones.

Market Pull

The term 'Market Pull', refers to the need/requirement for a new product or a solution to a problem, which comes from the market place. The need is identified by potential customers or market research. A product or a range of products are developed, to solve the original need.
Market pull sometimes starts with potential customers asking for improvements to existing products. Focus groups are often central to this, when testing a concept design or an existing product.

Eg. Digital camera

The digital camera. Twenty years ago, there was a 'market' requirement for a camera that could take endless photographs, that could be viewed almost immediately. Market pull (market need) eventually led to electronics companies developing digital cameras, once miniature digital storage, processing power and improved battery performance was available. Market pull ensured that photo editing software also developed, in parallel with the development of digital camera technology.

Attributes of creative person

1. **Imagination:** Creative individuals often have vivid imaginations and can envision possibilities beyond the ordinary.
2. **Curiosity:** They are naturally curious and eager to explore new ideas, concepts, and experiences.
3. **Open-mindedness:** Creativity thrives on being open to different perspectives, allowing for novel connections and ideas.
4. **Persistence:** Creativity often involves facing challenges and setbacks, so perseverance is crucial in pursuing new ideas to fruition.
5. **Flexibility:** Being adaptable and willing to consider alternative solutions or approaches is important in creative endeavours.
6. **Risk-taking:** Creativity often involves taking risks by exploring unconventional ideas or methods.
7. **Passion:** Creative individuals are often deeply passionate about their work, which drives them to overcome obstacles and keep exploring.

8. **Playfulness:** A sense of playfulness and the ability to experiment without fear of failure can lead to innovative ideas.
9. **Observant:** They notice details and patterns that others may overlook, which can inspire new ideas and insights.
10. **Communication:** Effectively conveying their ideas and visions to others is crucial for bringing creative projects to life.

GROUP CREATIVITY / CREATIVE THINKING IN GROUPS

A creative group is six to twelve people working together to clarify a situation, solve a problem or generate new ideas for later refinement. Group creativity refers to the collective ability of a team or a group of individuals to generate innovative ideas, solutions, or outcomes through collaborative efforts. It involves leveraging the diverse perspectives, skills, and experiences of team members to foster creativity and achieve creative breakthroughs that might not be attainable by individuals working alone. There are different activities that project managers can organize to identify ideas for the project to meet the requirements. Below mentioned are the different types of techniques that can enhance creativity and decision-making within the group.

Brainstorming

It is a group creativity technique in which members are allowed to generate as many ideas/requirements as possible without criticism. The brainstorming technique does not prioritize the ideas. In this technique, the participants are safe to present their very own creative ideas even though some ideas are unrealistic. During the process, all the generated ideas/requirements are recorded without any assessments. Additionally, a productive brainstorming session triggers one idea from another, enabling the team members to spot the connections between the ideas.

The Nominal Group Technique

It is a technique for small group discussion in which ideas and requirements are ranked and prioritized by all the team members of the group after generating the list of requirements. It enhances the brainstorming with a voting process that is used to rank the most useful ideas for further prioritization.

Mind Mapping

It is used through individual sessions with each team member. The project manager then consolidates the ideas into a single map to create a common ground of understanding.